

Foundations of Machine Learning

CS3390/CS5590

Assignment 1

Submission Deadline: October 8, 2026

Total: 150 Marks

Please read the following instructions before starting the assignment:

- Submit theoretical answers as a **PDF** and programming solutions as an **IPYNB** file.
 - Use the naming convention: `A1_<roll-number-member1>_<roll-number-member2>.pdf` and `A1_<roll-number-member1>_<roll-number-member2>.ipynb`.
 - Students are expected to maintain honesty and academic integrity throughout the assignment. Plagiarized work may be awarded an **F grade**.
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1. Mess Food Completion-Time Prediction

[The data collection should be carried out by two participants working independently over the same two-week period]

Please collect data on how long you will take to finish the food in the mess (including waiting time) for breakfast, lunch, tea, dinner, etc. for the next 2 weeks. The data should be collected by considering two people working on the data collection.

Prepare the data by considering various features like day, time, holiday, category (breakfast, lunch, tea, dinner, etc.), person, and other features that you think could be helpful for accurate prediction of the time you will take to finish the food.

- (a) Prepare the data and perform an exploratory data analysis, prepare a plot of the time taken against different days for each of meal type.
- (b) Due to the nature of the output, we know that linear regression may not be suitable for our problem. Suggest a model to fit the data and learn parameters using Maximum Likelihood Estimation using 80% of the data. Code the model and learn the parameters from the data. Analyze the parameter values and figure which parameter is important for prediction.
- (c) Use the learned model to do prediction. For prediction, you may consider 20% of the collected data. Provide an appropriate evaluation metric and compare the results of the prediction of your model with a linear regression model (you may use existing packages like scikit-learn for linear regression).

Provide explanations on the choice of your features and model. Code has to be in Python with sparse use of packages. Provide a Readme file explaining how to run your code and with proper comments.

Marks : 10 (a) + 15 (b) + 5 (c) = 30 + 10 (readable code) = 40

2. Dimensionality Reduction using PCA on MNIST

Use the **MNIST handwritten digit dataset**, which contains 28×28 grayscale images represented as 784-dimensional vectors. The dataset contains 60,000 training and 10,000 test images. Normalize the pixel values to $[0, 1]$.

Dataset: <https://www.kaggle.com/datasets/hojjatk/mnist-dataset/data>

PCA reference: <https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>

(a) PCA: Theory

Consider the covariance matrix

$$C = \begin{bmatrix} 0.25 & 0.20 \\ 0.20 & 0.25 \end{bmatrix}.$$

Find its eigenvalues and corresponding normalized eigenvectors. Identify the first principal component and calculate the percentage of total variance explained by it. Briefly explain how an input $x \in \mathbb{R}^{784}$ is projected onto a k -dimensional PCA space.

[Handwritten derivation: Show all intermediate mathematical steps and clearly state the final results.]

(b) 2D Representation of MNIST: Practical

Apply PCA to the MNIST training data and obtain a two-dimensional representation using the first two principal components.

- (i) Report the variance explained by PC1, PC2, and their cumulative variance.
- (ii) Project the MNIST test images onto the same two-dimensional PCA space and plot PC1 versus PC2 using different markers/labels for the ten digit classes (0–9).
- (iii) Select a few samples from each digit class and represent them in the 2D space, and discuss your observations.
- (iv) Reconstruct the selected images using only the first two principal components. Display the reconstructed images alongside the originals and comment briefly on the information retained and lost.

(c) Analysis and Interpretation

Based on your PCA results and visualizations:

- (i) Identify digit classes that are relatively well separated and those that show significant overlap in the 2D representation. Give possible reasons for the observed overlap.
- (ii) Explain why the first two principal components can still be useful for visualization even if they do not capture most of the total variance.
- (iii) Discuss what information is lost when reducing the original 784-dimensional representation to two dimensions. Support your discussion using the explained-variance and reconstruction results.

Marks: 5 (a) + 16 (b) + 9 (c) = 30

3. Read *Regression Models for Ordinal Data* by Peter McCullagh by Peter McCullagh (please find attached).

- (a) Provide a brief summary of the paper. Explain how the likelihood and odds ratio for ordinal regression is different from multi-class classification. How is it different from the regression problems?
- (b) Explain and derive the parameter estimation technique for the ordinal likelihood described in the paper.
- (c) Consider the *Wine Quality dataset*: <https://archive.ics.uci.edu/dataset/186/wine+quality> Develop an ordinal regression model in Python to predict the rating of the wine based on the input features. You may consider 20% of the data for testing purpose. Also, propose an appropriate evaluation metric and compare the result with a linear regression model. You may use existing ordinal regression and linear regression packages.

Marks: 7.5 (a) + 7.5 (b) + 15 (c) = 30 + 10 (readable code) = 40

4.[Handwritten derivations] The method of ordinary least squares assumes that there is constant variance in the errors, which is called homoscedasticity. The method of weighted least squares can be used when the ordinary least squares assumption of constant variance in the errors is violated (which is called heteroscedasticity).

- (a) Derive the expression of likelihood and prior for a heteroscedastic setting for a single data point with input x_n and output t_n .

- (b) Provide the expression for the objective function that you will consider for the ML and MAP estimation of the parameters considering a data set of size N .
- (c) Show that the ML objective results in each data point t_n being associated with a weighting factor $r_n > 0$, such that the sum-of-squares error function becomes

$$E_D(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N r_n \{t_n - \mathbf{w}^T \phi(x_n)\}^2.$$

Find an expression for $\mathbf{w}$ that minimizes this error function.

Marks: 5 (a) + 10 (b) + 5 (c) = 20

5.[Handwritten derivations] For logistic regression, there is no longer a closed-form solution, due to the nonlinearity of the logistic sigmoid function. The error function can be minimized by an efficient iterative technique based on the Newton–Raphson iterative optimization scheme.

- (a) Provide the expressions for the gradient, Hessian, and update equations for the Newton–Raphson optimization technique used to obtain the parameters in the logistic regression model. Provide an algorithm describing the methodology.
- (b) Show that the Newton–Raphson update scheme is related to the weighted least squares problem described in Question 4(c) and explain why it is called the Iteratively Reweighted Least Squares method.
- (c) Show that the error function of logistic regression is a convex function of $\mathbf{w}$ and hence has a unique minimum with the help of the Hessian matrix.

Marks: 10 (a) + 5 (b) + 5 (c) = 20